THE CAFÉ DOESN'T END AT THE COUNTER.

Coffee on QR turns café visits into ongoing customer relationships.



EXISTING COFFEE ON QR PRODUCT

QR customer access
Connect customers to a café.

Digital menu
Browse items and prices.

Loyalty and rewards
Encourage repeat visits.

Customer/CRM records
Understand returning customers.

Retention and WhatsApp engagement
Bring customers back.

Ordering-related workflows and analytics
Support interactions and track performance.

FROM QR ENGAGEMENT TO A SMARTER CAFÉ EXPERIENCE.

Coffee on QR | AI FOODTECH

NMIMS ADAPPT 5.0



Good Food Deserves a longer story.

WE ALREADY KNOW THE CAFÉ. NOW WE CAN HELP IT ACT EARLIER.

We're extending an existing café platform into AI-powered food intelligence.

THE OPERATIONAL PAIN

REAL CHALLENGES FACED BY CAFÉ OWNERS



1 UNSOLD INVENTORY

Food remains unsold at closing due to uncertain demand.



2 INCOMPLETE MONITORING

Temperature logs may miss cumulative time-temperature exposure.



3 MISSED CUSTOMER OPPORTUNITIES

Customers struggle to discover the right available dish or relevant offer.

! The cost is not just wasted food. It is missed sales, manual checks, and decisions made too late.

OUR STARTING ADVANTAGE

EXISTING CAFÉ NETWORK → CUSTOMER INTERACTIONS → PERSONALIZATION → AI FOOD INTELLIGENCE

✓ We can test inside the Coffee on QR café network first, instead of starting as a city-wide marketplace.

OUR TRACTION (LIVE PRODUCT)

04 CAFÉS
Existing café presence

₹50,000 REVENUE GENERATED
Real-world commercial traction

100+ CREATORS
Creator network built

10+ WORKSHOPS
Real-world community activity

NEXT EXPANSION → **BANDRA, MUMBAI**

COFFEE ON QR — EXISTING PRODUCT



REAL PRODUCT FOUNDATION. REAL TRACTION. THE NEXT AI FOODTECH LAYER.

ONE BATCH. ONE DECISION LOOP.

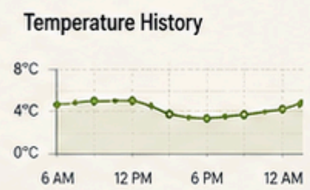
MONITOR CONDITIONS → PREDICT DEMAND → OFFER ONLY WHEN ELIGIBLE.

QUALITY GATES THE OFFER. CUSTOMER PREFERENCES TARGET IT.

1 MONITOR

In-house sensor captures time-stamped temperature history.





A cumulative time-temperature model tracks exposure over time. A single current reading is not the whole story.

2 FORECAST

Use sales history, current inventory and predicted demand.



Predict whether stock is likely to sell before the permitted sale window closes.

3 SAFETY / ELIGIBILITY GATE

Check sensor data and food handling rules.

- **Within configured limits** — Eligible for offer (with manager approval).
- **Warning / review** — Staff to check and confirm.
- **Deviation or permitted window ended** — Do not offer. Keep for internal review / discard as per café's procedure.



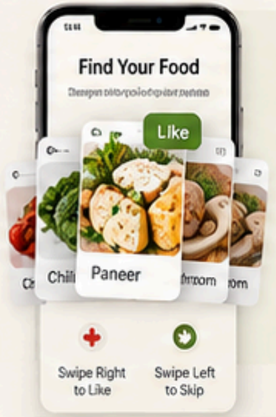
FOOD BATCH

+ REMAINING ELIGIBLE SALE WINDOW.

Prepared 10:00 AM | Sandwich (Veg) | Eligible till 8:00 PM

4 MATCH

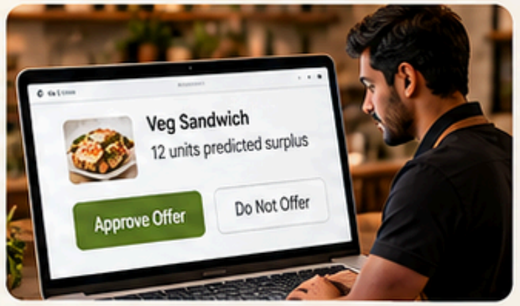
Use customer preferences to find the right audience.



- Ingredient swipes and dietary preferences
- Order history
- Available menu items at nearby cafés
- Budget, cuisine and taste profile

5 APPROVE + OFFER

Manager approves and sends offer to the right customers.



Send via opted-in WhatsApp / in-app

Customers reserve through Coffee on QR

Eligible for offer

Not eligible (Blocked)

No offers are sent for items with a safety concern or whose permitted window has ended.

6 LEARN

Use outcomes to improve future decisions.

Completed orders | Unsold units | Alert response | Forecast error

Improve future demand prediction, recommendations and operations.

SENSOR DATA. DEMAND PREDICTION. CUSTOMER MATCHING. MANAGER ACTION. CONTINUOUS LEARNING.

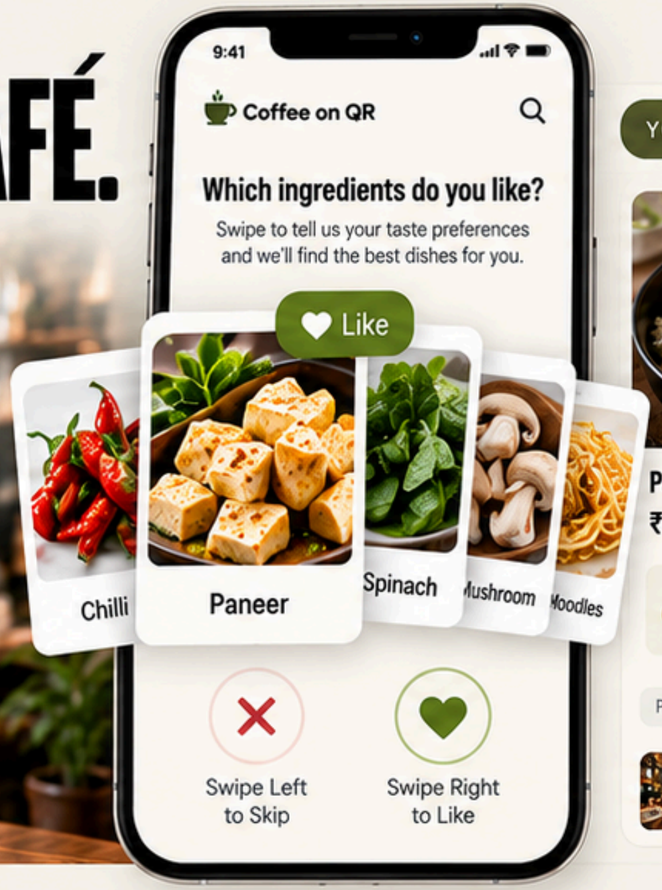
Coffee on QR | AI FOODTECH

NMIMS ADAPPT 5.0

DON'T JUST FIND A CAFÉ. FIND YOUR FOOD.

PERSONALIZED DISCOVERY INSIDE THE COFFEE ON QR NETWORK FIRST.





YOUR PERSONALIZED FOOD MATCHES

Based on your ingredients, goals and preferences



Paneer Tikka Bowl
₹ 220 Available nearby

Matches your spicy vegetarian preferences.

Paneer | Capsicum | Tandoori



Available at **The Brew Room Café**



Grilled Chicken Bowl
₹ 260 Available nearby

Protein-rich meal
Fits your selected protein goal.

Chicken | Quinoa | Greens



Available at **Urban Bean Café**



Creamy Mushroom Pasta
₹ 240 Available nearby

Contains peanuts
Ask staff to confirm a suitable alternative.

Mushroom | Pasta | Peanut



Available at **Café 47**

HOW IT WORKS

1. **SWIPE**
Like or skip ingredients you prefer or want to avoid.

2. **MATCH**
Personalize by taste, budget, cuisine and declared diet/nutrition preferences.

3. **DISCOVER**
Find and rank dishes from participating cafés using live menu availability.

4. **LEARN**
Improve recommendations using consented preferences, feedback and order history.

KEY FEATURES

LIVE AVAILABILITY | SMART FILTERS | TASTE MEMORY | CLEAR INGREDIENT INFORMATION

LESS DECISION TIME · MORE RELEVANT ORDERS · BETTER REPEAT VISITS

REVENUE MODEL:
COMMISSION ON COMPLETED ORDERS + CLEARLY LABELLED PROMOTED PLACEMENTS.

A TEMPERATURE SNAPSHOT ISN'T THE WHOLE STORY.

THERMAL DEBT — TRACK CUMULATIVE TIME-TEMPERATURE EXPOSURE, BATCH BY BATCH.

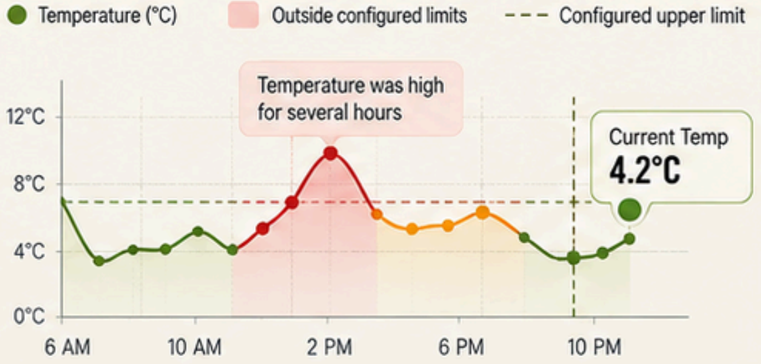
1 OUR IN-HOUSE PRODUCT SENSOR

Continuously records temperature and time.



2 SNAPSHOT vs THERMAL DEBT

A single reading can be misleading.
We track the full temperature history.



Snapshot alert

4.2°C

Reports the temperature right now.

Thermal Debt prototype

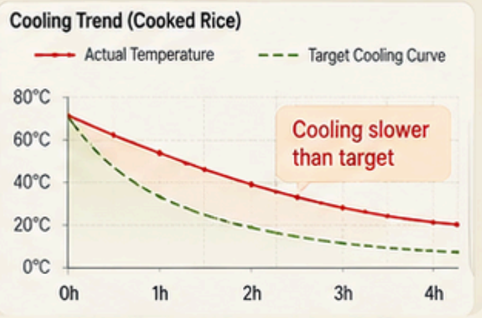
Remaining eligible window
~ 3 hours

Considers recorded history and time spent outside configured conditions using a transparent model.

3 EARLY COOLING WARNING

Predicts if cooked food is cooling too slowly.





Cooling trend may miss the configured target.
Split into shallow trays and follow the approved procedure.

4 STAFF-ONLY ALERT FLOW

Helps teams act early and document decisions.

PROBE / SENSOR → TIME-STAMPED HISTORY → RULES + EXPOSURE MODEL → STAFF ALERT → HUMAN INSPECTION / DOCUMENTED ACTION

5 CUSTOMER-FACING VIEW (SIMPLIFIED)

Customers see that storage is monitored, not internal risk states.



Storage monitored
Updated 2:15 PM

Decision-support model; parameters require food-safety validation.
Sensor readings indicate monitored conditions and potential risk; they do not prove that food is safe or rotten.

LOGS + EARLY ALERTS + BATCH HISTORY; SENSOR OUTAGES ARE FLAGGED AS UNKNOWN, NEVER GREEN.

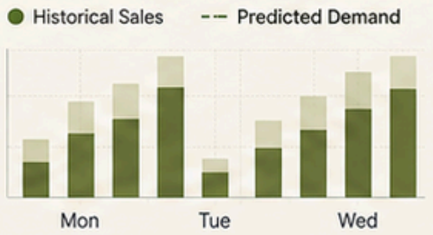
ACT BEFORE SURPLUS. BECOMES WASTE.

FORECAST EARLY. TARGET THE RIGHT CUSTOMER. RECOVER VALUE.

1 PREDICT

Analyse recent sales, current inventory and day/time patterns to estimate upcoming demand.

Demand Forecast (Next 3 Days)



Helps identify items that may remain unsold.

2 FLAG

Identify a product likely to remain unsold before the permitted sale window closes.



12 Sandwiches
Potential surplus

Based on predicted demand and current inventory.

Prepared: Today, 11:00 AM

Estimated sale window: Till 8:00 PM

3 GATE

Confirm the item remains eligible for sale under the café's food-safety process.

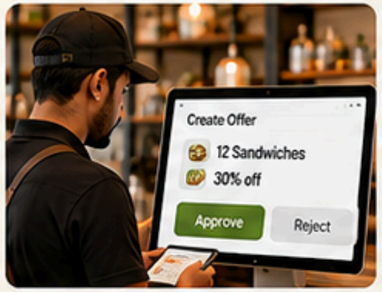
Eligible for Offer
Within configured conditions.

Not eligible
Safety concern or permitted window ended. Do not promote.

Sensor data unavailable
Review before any offer.

4 APPROVE

Café manager approves the offer via dashboard or WhatsApp (if configured).

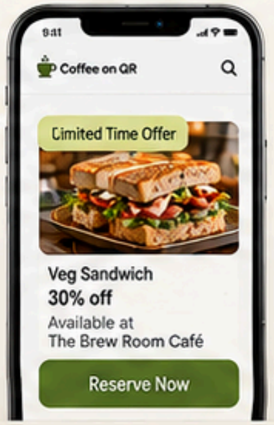


OR

Reply YES to publish the offer to matched customers.

5 MATCH + MEASURE

Send the approved offer to opted-in customers whose preferences fit. Track reservations, sales and remaining unsold units.



Relevant customers receive the offer (WhatsApp / in-app).

Customers reserve and pick up.

Track sale, discount cost and remaining unsold units.

COMPLETED ORDERS, UNSOLD UNITS AND FEEDBACK IMPROVE FUTURE FORECASTS AND CUSTOMER TARGETING.

BUSINESS MODEL

CLEAR BUYERS. MULTIPLE REVENUE PATHS.

Café owner / operations lead
Monthly condition-monitoring and forecasting plan.
Monthly plan: ₹[validated price]

In-house sensor
Sale or rental, priced from a real supplier quote.
Sensor kit: ₹[supplier quote]

Completed eligible orders
Transaction commission.
Order commission: [tested rate]%

Multi-outlet café groups
Central monitoring and analytics.
Pilot payback: calculate from measured results

COMPETITION / POSITIONING

DIFFERENT PLAYERS SOLVE PARTS OF THE PROBLEM.

Discovery
Zomato / Swiggy / Magicpin
(and similar platforms)

Surplus
Too Good To Go model
(and similar platforms)

Monitoring
Temperature-log vendors
(and similar providers)

OUR EDGE: COFFEE ON QR'S EXISTING CAFÉ RELATIONSHIPS + FIRST-PARTY CUSTOMER PREFERENCES + SAFETY-GATED, TARGETED OFFERS.

PILOT / NEXT STEP

START WITH A FOCUSED PILOT INSIDE OUR NETWORK.

30-DAY PILOT: [3–5] cafés
Establish baseline and measure: forecast error, unsold units, offer conversion, sensor uptime and net recovered revenue.

Café count to be confirmed based on interest from existing partners.

MEASURE FIRST. PROVE VALUE. SCALE WHAT WORKS.

Coffee on QR | AI FOODTECH

TARGET ARCHITECTURE — NMIMS ADAPPT 5.0

ONE CAFÉ PLATFORM. ONE CONNECTED FOOD INTELLIGENCE SYSTEM.

From QR customer engagement to AI-powered discovery, food-condition monitoring and waste recovery.

1 CHANNELS
Customer & Café Inputs



Customer Mobile App
Food discovery and preferences



Café Staff Dashboard
Inventory and action approvals

QR Entry Point
Existing Coffee on QR experience



In-house Product Sensor
Temperature and storage-condition readings

2 INGESTION LAYER
Data Entry

Customer / API Requests
Preferences, menus and orders

Sensor Ingestion API
Timestamped sensor readings

Menu Import
Photo / PDF extraction with owner confirmation

Sales & Inventory Import
POS or CSV data where available

3 APPLICATION LAYER
Product Backend

Customer & Preference Service
Profiles, preferences and personalization data

Café and Menu Service
Menus, outlets, availability

Sensor & Batch Monitoring Service
Temperature history and batch tracking

Inventory, Orders & Offer Management
Orders, surplus, offers and fiulfillment

API Authentication and Access Control
Secure access for apps, dashboards and integrations

4 AI DECISION LAYER
Food Intelligence

Food Discovery Engine
Ranks available dishes using ingredient preferences, budget, dietary preferences and declared nutrition information.

Thermal Debt / Condition Engine
Tracks time-temperature history, configured limits and cumulative exposure to flag batches requiring review.

Demand Forecasting Engine
Uses sales history, inventory, day/time patterns and context to estimate demand and potential surplus.

Offer Targeting Engine
Matches approved, eligible offers to interested customers using their preferences and opt-in status.

5 DOMAIN TOOLS
Actions

Menu & Availability Lookup
Check item details and current availability

Batch Condition History
View sensor history and exposure summary

Inventory Forecast
Identify potential surplus items

Manager Offer Approval
Approve discount / offer via dashboard or WhatsApp

Customer Notification
Send to opted-in customers

Order Reservation & Outcome Tracking
Track redemptions, sales and remaining units

6 DATA LAYER
System of Record

PostgreSQL / Managed Relational Database

- Customers & Preferences
- Cafés & Menus
- Orders & Sales History
- Sensor Readings & Batch History
- Inventory & Forecasts
- Offers & Notifications
- Analytics & Audit Logs

7 ASYNC LAYER
Event-Driven Processing

Sensor Alert Jobs
Process new sensor data and threshold breaches

Scheduled Demand Forecasts
Run periodic forecasts

WhatsApp / In-App Notifications
Send approved offers and staff alerts

Retries & Failure Handling
Ensure reliable delivery

Background Analytics Jobs
Generate reports and model updates

8 EXTERNAL SYSTEMS
Integrations

WhatsApp Business API
Staff alerts and approved offer notifications

Sensor Gateway
Device communication (REST / MQTT)

POS / CSV Sources
Sales and inventory imports

Payment or Ordering Provider
Order processing where integrated

CLOSED-LOOP BUSINESS WORKFLOW

SENSOR HISTORY → Continuous readings and time-temperature data

CONDITION & ELIGIBILITY GATE → Check cumulative exposure and business rules

DEMAND FORECAST → Estimate likely unsold items

MANAGER APPROVAL → Approve eligible offer

PERSONALIZED OFFER → Send to the right customers

CUSTOMER RESERVATION → Order and pick up

SALES & INVENTORY UPDATE → Record outcomes

A batch flagged for a safety concern is blocked from promotional offers until appropriately assessed.

Completed orders and remaining inventory improve future forecasts and customer targeting.

DEPLOYMENT & OPERATIONS (Proposed Hackathon Architecture)

Frontend Hosting	API / Backend Services	Managed Database	Sensor Connectivity	Monitoring & Logs	Secrets & Access Control
Next.js + TypeScript	Python + FastAPI	PostgreSQL / Supabase	REST / MQTT	Logging, alerting and observability	Environment management and access control

DISCOVER BETTER. MONITOR SMARTER. RECOVER VALUE.